

RISHABH ARVIND KARPE

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OBJECTIVE

Aspiring IT professional with hands-on experience in system setup, troubleshooting, and technical documentation. Skilled in configuring devices, maintaining infrastructure, and providing end-user support. Seeking to contribute as an MDS Intern at Mondelēz International by supporting IT operations, system maintenance, and knowledge-sharing initiatives.

EDUCATION

Bachelor of Technology, Electronics & Computer Engineering 2022 - Present

KJ Somaiya College of Engineering, Mumbai, India

Relevant Coursework: Networks, Data Structures & Algorithms, Embedded Systems, IoT Architectures

SKILLS

IT & Infrastructure	System setup & configuration (Windows/Linux, Raspberry Pi, Arduino); Network basics (LAN/Wi-Fi troubleshooting, MQTT/HTTP, VNC/VMware); System maintenance (updates, backups, monitoring tools); Documentation & user training guides
Technical Tools	Visual Studio Code, VMware, RealVNC, AutoCAD , TIA Portal, MS Office, GitHub
Soft Skills	Problem-solving, adaptability, teamwork, verbal & written communication, self-directed learning, attention to detail

EXPERIENCE

Intern July 2025 – September 2025
Smart Aerofarm Project (KJ Somaiya College of Engineering) *Mumbai, India*

- Worked on a vertical farming automation system integrating sensors and actuators for optimal plant growth.
- Configured Raspberry Pi with IoT-based monitoring (temperature, humidity, and soil parameters) for real-time data collection.
- Developed control logic to automate irrigation and lighting based on environmental thresholds.
- Collaborated with a cross-functional team to implement data visualization and remote monitoring via cloud dashboards.

Creative Head Jun 2024 – Present
Ecesa Council *Mumbai, India*

- Coordinated IT/design support for technical events, including posters, training materials, and digital assets.
- Managed end-to-end digital campaigns to boost outreach and ensured consistent delivery.

Intern May 2024 – June 2024
Improsys *Pune, India*

- Configured Raspberry Pi and integrated sensors for monitoring and automation, simulating IT system setup.
- Supported infrastructure maintenance by optimizing power consumption through automated control logic.
- Prepared technical documentation and presented system outcomes to the team.

Intern Jan 2024
Basic Communication for Paralyzed Patients (KJ Somaiya College of Engineering) *Mumbai, India*

- Developed an assistive communication system by configuring hardware and software modules.
- Created setup and usage documentation for end-users.
- Conducted troubleshooting and system testing across scenarios to ensure stability.

PROJECTS

IoT-Based Farm Automation System

Jan 2025 – Feb 2025

- Automated irrigation control by integrating DHT11 and soil moisture sensors with Raspberry Pi and Firebase for real-time monitoring and control.
- Optimized water usage through threshold-based triggers and scheduled reports.
- **Tech:** Raspberry Pi, Arduino, Python, Firebase, DHT11, Soil Moisture Sensor, Motor Driver
- **Transferable Skills:** Critical thinking, time management, problem-solving, innovation

Home Electricity Saving System

May 2024 – Jun 2024

- Built a power-saving solution by detecting room occupancy with YOLOv4 and OpenCV on Raspberry Pi, automatically toggling lights/fans.
- Demonstrated potential to reduce wasteful energy consumption.
- **Tech:** Python, OpenCV, YOLOv4, Raspberry Pi
- **Transferable Skills:** Numeracy, attention to detail, problem-solving, commercial awareness

Basic Communication for Paralyzed Patients

Jan 2024

- Built an assistive communication prototype by tracking eye movements with YOLOv4 on Raspberry Pi and mapping gaze to voice commands via a speaker.
- Enabled hands-free communication through pre-defined voice outputs such as “Need help” or “Bring water.”
- **Tech:** Raspberry Pi, Python, OpenCV, YOLOv4
- **Transferable Skills:** Innovation, empathy-driven design, IT/Design fluency, verbal communication

V.I.S.T.A AI – Vision Integrated Smart Assistant

Jun 2025 – Ongoing

- Created a multimodal AI assistant with speech recognition (Vosk), wake word detection (Porcupine), object detection (YOLOv4), and hand gesture recognition (MediaPipe).
- Integrated with FastAPI, Streamlit, and SocketIO for real-time interaction and modular control.
- **Tech:** Python, FastAPI, Streamlit, SocketIO, Vosk, MediaPipe, Porcupine, YOLOv4
- **Transferable Skills:** Multimodal AI, real-time interaction, system integration, open-source fluency

TaskPlanner-AI

Jun 2025

- Developed a LangGraph-based AI system using Mistral 7B (via Ollama) that plans, executes, and evaluates user-defined goals using agent-based task breakdown.
- Deployed with a clean Streamlit UI and integrated FAISS for persistent memory.
- **Tech:** LangGraph, Streamlit, Ollama, Mistral 7B, FAISS
- **Transferable Skills:** Systems thinking, modular design, prompt engineering, orchestration

Semantic PDF Q&A Bot

Jun 2025

- Built a semantic search-powered assistant that allows users to ask natural-language questions from PDFs.
- Used sentence-transformer embeddings with FAISS for fast and accurate contextual retrieval from large documents.
- **Tech:** Python, FAISS, SentenceTransformers, Streamlit
- **Transferable Skills:** Semantic search, NLP, user-centric design, data preprocessing

ACHIEVEMENTS

- Won PAN IIT Hackathon 2025: Out of 17,000+ registrations, our team reached the top 72 finalists and emerged as winners.

EXTRA-CURRICULAR ACTIVITIES

- Designed and shared promotional content for college technical events, boosting awareness and participation.
- Regularly assist peers in troubleshooting IoT/IT setups and exploring new tech blogs, tutorials, and tools.
- Actively participate in hackathons and coding challenges, honing problem-solving and rapid prototyping skills.